PATLEGAR MANJUNATHA

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THE JOURNEY & PROFESSIONAL PHILOSOPHY

Ambitious entry-level Full Stack Machine Learning Engineer focused on the complete lifecycle of artificial intelligence. While traditional data science focuses on theoretical model architectures, the core philosophy here is bridging the gap between Jupyter Notebooks and production-grade cloud environments. Specializes in taking raw ML experiments and wrapping them in robust, scalable, and automated infrastructure (CI/CD, containerization, observability) to build autonomous systems that solve real-world enterprise problems.

PROFESSIONAL STRENGTHS

- **End-to-End MLOps Engineering:** Highly capable in managing the entire stack, from data ingestion to model deployment using Docker, Kubernetes, DVC, and MLflow.
- **GenAI Orchestration:** Skilled in building autonomous agents and RAG pipelines using LangChain, LangSmith, and Knowledge Graphs (Neo4j).
- **Systems Architecture:** Strong ability to design decoupled microservices, separating prediction serving APIs from heavy model-training workloads using FastAPI.
- **Pragmatic Problem Solving:** Prioritizes engineering maturity, system reliability (drift detection, automated testing), and functional capability over industry buzzwords.

AREAS FOR GROWTH & CONTINUOUS LEARNING

- **MVP vs. Enterprise Scaling:** Acknowledged tendency to overengineer solutions (e.g., applying Kubernetes to simple datasets) for the sake of learning. Actively working on balancing Minimum Viable Product (MVP) agility with full-scale production architecture.
- **Full-Stack UI Expansion:** Historically focused on backend and AI infrastructure. Currently pushing boundaries by building modern frontend applications to operate as a completely autonomous Full-Stack AI Engineer.

ACADEMIC INTERNSHIPS

Academic Machine Learning Intern | Acenaar Technologies | Nov 2025 – Mar 2026 | Kurnool, India (On-Site)

- What was done: Developed Python processing pipelines utilizing Pandas to ingest, clean, and standardize raw JSON telemetry streams from educational IoT/AI hardware kits.
- Architecture: Architected lightweight REST API endpoints via FastAPI to expose processed sensor data to local demonstration dashboards.
- Impact: Implemented local SQLite databases to log API requests and hardware states, ensuring reliable audit trails and smooth product demonstrations.

Academic Machine Learning Intern | Skill Dzire | May 2025 – July 2025 | Remote

- Engaged in hands-on academic training and practical implementation of Machine Learning algorithms and data preprocessing techniques.
- [Add specific project or core responsibility here]

PROJECT PORTFOLIO

Enterprise MLOps Framework (Project IRIS) | FastAPI, Docker, Kubernetes, AWS, MLflow

- The Problem: Proving that the ML model is only 5% of the challenge, while the remaining 95% requires robust infrastructure, tracking, and serving architecture.
- The Solution: An intentionally overengineered pipeline built around a simple dataset to showcase advanced production MLOps practices.
- The Architecture: A decoupled microservice ecosystem (Training Worker vs. Prediction API) that uses shared Docker volumes for dynamic artifact loading, allowing the FastAPI server to hot-swap models with zero downtime.

SaaS Customer Retention Engine | Python, MLflow, DVC, MinIO, FastAPI

- The Problem: Businesses lack the ability to proactively identify and flag subscription flight risks before the user churns.
- The Solution: An end-to-end MLOps batch inference pipeline that scores customer churn probability and dispatches automated SMTP reports.
- The Architecture: Utilizes DVC and MinIO (S3-compatible) for version control, automated MLflow retraining loops, and Pytest/Pydantic schemas integrated into the CI/CD workflow for strict data reliability.

Conversational Next Word Predictor | TensorFlow, Deep Learning, NLP, Python

- The Problem: Standard text generators trained on Wikipedia data output formal, robotic text that fails to capture natural conversational flow.
- The Solution & Architecture: A custom Stacked LSTM Deep Learning model trained on the DailyDialog dataset. Replaced greedy search with Temperature Sampling (Top-K/Random Choice) to introduce variance and prevent repetitive output loops.

CURRENT FOCUS & ACTIVE BUILDS

Project Veda (Edge AI & Computer Vision) | OpenCV, ChromaDB, Groq API, Edge AI

- The Scope & Problem: Cloud-dependent AI suffers from latency and privacy concerns during real-time human interaction. The goal is to build a local-first virtual human interface.
- The Execution: Utilizing OpenCV for real-time facial and object recognition (Computer Vision), ChromaDB for contextual long-term memory, and the Groq API for ultra-low latency GenAI inference without heavy cloud dependencies.

Nexus AI (GenAI & Knowledge Graphs) | LangChain, Neo4j, Python, FastAPI

- The Scope & Problem: Traditional ATS platforms fail at semantic understanding, rejecting highly qualified talent due to rigid keyword string matching.

- The Execution: Architecting an autonomous GenAI recruitment agent. Using LangChain to parse unstructured PDF resumes and Neo4j (Knowledge Graph) to infer hidden skill relationships to calculate dynamic capability scores.

SizeSnap (Client-Side Recommendation Engine) | JavaScript, Manifest V3, DOM API

- The Scope & Problem: Shoppers experience high friction cross-referencing static size charts for every individual e-commerce product.
- The Execution: Engineering a highly optimized, zero-server footprint Chrome extension that calculates weighted fit-scores and displays personalized recommendations directly on Ajio/Myntra product cards.

EDUCATION

B.Tech in Computer Science and Artificial Intelligence | St. Johns College of Engineering and Technology (JNTUA) | Adoni, India
Graduated: April 2026 | CGPA: 7.44 | Core Coursework: OS, DBMS, Computer Networks, Advanced Python

CERTIFICATIONS

Machine Learning Crash Course (Google), SQL for Database Design (Scaler), Data Science Certification (Cisco), Python Essentials I & II (Cisco)